Maternal Health Fitband

Proactive, Continuous Care for Expectant Mothers

The Gap in Maternal Care

- Pregnant women often experience periodic discomfort and warning signs of complications—such as preeclampsia—that go completely unrecorded between scheduled clinic visits.
- Without continuous tracking of vital parameters at home, doctors miss critical data, and mothers lack personalized, data-driven guidance for their daily diet, sleep, and exercise.

The Clinical Need

- Expectant mothers lack continuous home monitoring for vital signs, leaving critical gaps in care between doctor visits. We must address this fundamental gap:
- **"A way to proactively detect and manage physiological warning signs in at-risk expectant mothers outside the clinic, to reduce the incidence of severe gestational complications."**

Need Scoping (Asking the Right Questions)

- a. Clinical Relevance
  - Hypertension are leading causes of maternal morbidity.
  - Delayed detection of vital fluctuations between clinic visits worsens outcomes.
- b. Stakeholder Perspective
  - Patients: want reassurance, comfortable tracking, and actionable daily guidance.
  - Doctors: need accurate, continuous vital data to adjust care plans without data overload.
  - Payers/Insurance: support preventative tech that reduces emergency hospitalizations.
- c. Market Potential
  - Rapidly expanding global maternal digital health market.
  - Growing demand for consumer-friendly wearables integrated with clinical care.

Need Scoping (Asking the Right Questions)

- d. Technical Feasibility
  - Approaches: PPG sensors for BP trends, accelerometers.
  - Challenges: Accuracy during pregnancy-induced edema.
- e. Regulatory & Reimbursement
  - Would require FDA/CE approval if providing diagnostic alerts or clinical decisions.
  - Reimbursement possible if the device demonstrates reduced long-term maternal complications.
- f. Competitive Landscape
  - Current solutions: Standard BP cuffs,generic fitness trackers.
  - Limitations: Standard medical devices are episodic and uncomfortable; generic trackers lack pregnancy-specific AI.

Our Solution: Maternal Health Fitband

- The fitband utilizes PPG to track blood pressure trends,oxygen saturation,heart rate alongside accelerometers for detailed sleep and activity analysis.
- Our integrated AI analyzes these physiological signals to infer health trends, delivering personalized recommendations for nutrition, sleep schedules, and exercise routines directly to the user.
- Designed for comfort and safety during pregnancy, the wearable seamlessly syncs with a smartphone app to log observations and alert care teams when vital thresholds are crossed.